

VENKAT DINESH PASUPULETI

Applied ML Scientist · AI Engineer · Production GenAI & RAG · MLOps

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Windsor / Toronto, ON, Canada | Valid Canadian Work Permit · No sponsorship required | Immediate Start

7+ Years | **90%** Fraud Precision | **~\$500K** Savings/yr | **50%** OCR Reduction | **DP-100** Certified

PROFESSIONAL SUMMARY

- Applied ML Scientist with 7+ years shipping production AI/ML systems across financial services and SaaS — from problem framing and model training through containerized deployment, monitoring, drift detection, and iteration. Not a notebook builder: I deploy, I monitor, I own the outcome.
- Currently leading AI engineering at DeskIQ.ai: built and deployed a full production AI SaaS — 42 modules, 557 FastAPI endpoints on GKE with autoscaling and CI/CD. Also shipped a Production RAG System (live: venkatdinesh-rag.hf.space) with hybrid retrieval (BM25 + FAISS), cross-encoder reranking, citation enforcement, streaming, and a CI-gated evaluation pipeline.
- 4+ years in regulated fintech at Vreedhi Financial Services: fraud detection at 90% precision saving ~\$500K/yr, Azure ML OCR pipeline cutting manual entry by 50%, XGBoost credit models with SHAP for compliance teams. This is where I learned that a model that can't be explained to an auditor isn't a model — it's a liability.
- Azure ML (DP-100 certified), GCP (GKE, Cloud Run, Vertex AI), Docker, Kubernetes, MLflow, GitHub Actions CI/CD. Clear communicator — translate trade-offs for engineers and compliance stakeholders alike.

TECHNICAL SKILLS

- GenAI & LLMs:** GPT-4.1, GPT-4o-mini, Claude Sonnet 4.5, Gemini, Llama 3.2, Mistral, RAG (Hybrid: BM25+FAISS+Reranker), Agentic Workflows, Prompt Engineering, Tool-Calling, LangChain, LangGraph, RAGAS, Streaming, Citation Enforcement
- Classical ML:** XGBoost, Random Forest, Extra Trees, SVM, Isolation Forest, TensorFlow, PyTorch, CNN (YOLOv3, Faster R-CNN), SHAP, Feature Engineering, Cross-Validation, ARIMA, Clustering
- Document Intelligence:** Azure AI Document Intelligence, OCR Pipelines, PDF Parsing (PyMuPDF), Table/Key-Value Extraction, Financial Statement Parsing, Named Entity Extraction
- Engineering:** Python 7+ years, FastAPI, REST/WebSocket APIs, Pydantic, async/await, Pytest, R (caret, ggplot2, tidyverse), SQL, MongoDB, PostgreSQL, Redis, FAISS, PySpark
- Cloud & MLOps:** Azure ML (DP-100), GCP (GKE, Cloud Run, Vertex AI, BigQuery), Docker, Kubernetes, MLflow, GitHub Actions CI/CD, Azure DevOps, Prometheus, Grafana, PSI Drift Detection, HuggingFace Spaces
- Governance:** JWT, OAuth 2.0, MFA/TOTP, RBAC (71 permissions), Audit Logging (365-day/7-year), PIPEDA-aware, OSFI Model Risk, SHAP Explainability for Compliance

PROFESSIONAL EXPERIENCE

Lead AI/ML Engineer — Production AI SaaS | Jay Analytix Inc (DeskIQ.ai) Aug 2025 – Present

Windsor, ON · Remote · GCP · Twilio · Claude Sonnet 4.5 · GPT-4.1

Sole backend engineer for DeskIQ.ai — production AI Voice Receptionist SaaS. 42 modules, 557 API endpoints on GKE. Telephony via Twilio; LLM inference via GPT-4.1 and Claude Sonnet 4.5.

- Agentic Tool System:** 8 custom tools (booking, availability, customer lookup) with JSON schemas + FastAPI execution endpoints. LLM selects and invokes tools mid-conversation. Zero-shot intent classification gate routes callers to KB retrieval vs tool execution.
- Production RAG System (Live — venkatdinesh-rag.hf.space):** Hybrid retrieval (BM25 + FAISS vector search) + cross-encoder reranking. Citation enforcement prevents hallucinations. Streaming responses. Per-query observability: latency, tokens, confidence. 50+ test suite with CI-gated evaluation pipeline. Dockerized, deployed on HuggingFace Spaces.
- LLM Cost Engineering:** Dual-prompt architecture — 462-line inbound / 19-line outbound. 95% token reduction on batch campaigns. Resolved production incident: concurrency limiting + prompt split cut p99 latency from 8s to 1.2s.
- Enterprise Security:** 71-permission RBAC enforced on every route, 365-day audit logging with 7-year retention, JWT + OAuth 2.0 + MFA/TOTP, human-in-the-loop gates for high-stakes AI actions. Maps to OSFI/PIPEDA requirements.
- MLOps:** GKE autoscaling (HPA min 2, max 20), GitHub Actions CI/CD (lint → test → Docker build → deploy → health check), MLflow model versioning, Prometheus + Grafana monitoring (p50/p95/p99 latency, per-call cost).

AI/ML Engineer (Intern) | Jay Analytix Inc

Feb 2025 – Aug 2025

Ontario, Canada · Remote

- RAG Chatbot:** Built intent-aware RAG chatbot: semantic search → dynamic SQL generation → Matplotlib/Plotly visualization. Accelerated reporting by 40%.

Lead Data Scientist — Fraud, OCR & Risk | *Dr. Reddy's Foundation (Vreedhi Financial Services)* Jun 2023 – Jan 2025
Hyderabad, India · On-site · Regulated Fintech · Azure ML
Led end-to-end ML for regulated lending fintech. Partnered directly with risk, compliance, and audit teams.

- **Fraud Detection — 90% Precision, ~\$500K/yr Savings:** YOLOv3 + Faster R-CNN (TensorFlow) detecting document tampering. Precision optimized over recall — false positives cost more than misses in lending. SHAP for compliance audit. Azure ML GPU training, CPU inference endpoint.
- **OCR Pipeline — 50% Manual Reduction:** Azure AI Document Intelligence for income statements, bank statements, payslips. Table extraction, key-value parsing, structured JSON output. PII-aware: encrypted storage, audit trail.
- **Loan Approval — XGBoost + SHAP:** ~70% accuracy, 5-fold stratified CV, per-prediction SHAP explanations for every denial. Azure ML managed endpoint with Azure DevOps CI/CD evaluation gate (precision ≥ 90% to promote).
- **R Statistical Modeling:** Regression, time series, segmentation in R (caret, ggplot2, dplyr). Prototyped in R, ported to Python for Azure ML deployment.

Data Scientist — Predictive Analytics | *Dr. Reddy's Foundation (Vreedhi Financial Services)* May 2021 – May 2023
Hyderabad, India · On-site

- **Customer Segmentation:** Extra Trees (82% accuracy, 100K+ records). ML-driven targeting → ~20% campaign engagement uplift.
- **Credit Risk:** XGBoost loan approval with SHAP interpretability. Azure ML endpoint + Power BI dashboards for compliance.

Data Analyst — Python, R, SQL & Analytics | *Vinfosoft Solutions PVT LTD* Dec 2018 – Apr 2021
Hyderabad, India · On-site

- **Forecasting & Automation:** ARIMA demand forecasting, 3x SQL optimization, automated Power BI/Tableau dashboards. 40% manual reporting reduction.

PROJECTS

Production RAG System — [Live: venkatdinesh-rag.hf.space](#)
Hybrid retrieval (BM25 + FAISS) + cross-encoder reranking + citation enforcement + streaming + CI-gated eval pipeline. Docker. 50+ tests. \$0/month.

- **Stack:** Python · FastAPI · BM25 · FAISS · Cross-encoder · GPT-4o-mini · Docker · HF Spaces

FinRoute AI — [SLM/LLM Routing for Financial Document Intelligence](#)
Routes tasks between SLM (Phi-3 Mini, <50ms) and LLM (Claude Sonnet 4.5) via 4-signal complexity classifier. 73% SLM-handled. 8x cost reduction. XGBoost fraud + Isolation Forest anomaly + SHAP.

- **Stack:** Python · FastAPI · Phi-3 Mini · Claude Sonnet 4.5 · XGBoost · FAISS · GCP Cloud Run · MLflow

EDUCATION

PG Diploma — Data Analytics <i>St. Clair College, Windsor, ON GPA: 3.25/4</i>	Jan 2025 – Apr 2026
PG Program — Data Science <i>INSOFE, India Grade: A Hackathon: 8/120 Teams</i>	Mar 2020 – Mar 2021
BE — Electronics & Communications <i>SRIT Engineering College, India</i>	2013 – 2017

CERTIFICATIONS

- Microsoft Certified: Azure Data Scientist Associate (DP-100)
- Prompt Engineering & Programming with OpenAI — Columbia+ (2026)
- Introduction to Responsible AI — Google Cloud (2025)
- Introduction to Large Language Models — Google / Coursera
- Best Performance Award (2022) · Ace of Initiative (2021) — Dr. Reddy's Foundation